

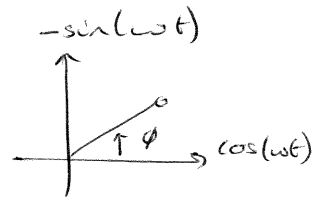
Chapter 9

Lecture 4:

* Phasors and impedance.

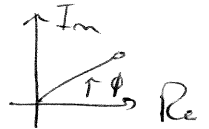
Time:

$$v(t) = V_m \cos(\omega t + \phi)$$



Phasor:

$$\bar{V} = V_m \angle \phi$$



→ Do not equate the two domains!

$$v(t) \neq \bar{V}$$

$$v(t) = \text{Re} [\bar{V} * e^{j\omega t}]$$

$$= \text{Re} [V_m \angle \phi * e^{j\omega t}]$$

→ Small letter: time → Real $v(t)$

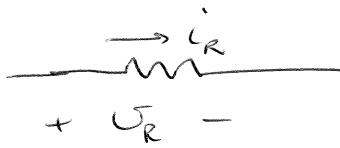
Capital letter: phasor → complex $\bar{V}$

Remember: phasor has size and direction.

* Relationship between phasors and R, L and C.

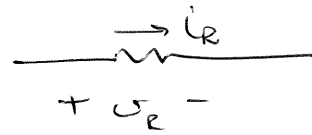
Time

R



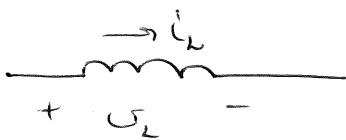
* $V_R = i_R \times R$

Phasor.

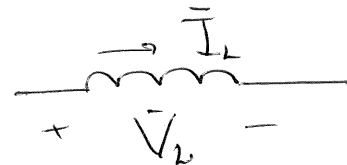


* $\bar{V}_R = \bar{I}_R \times R$

L

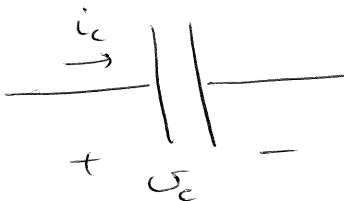


* $V_L = L \frac{di_L}{dt}$

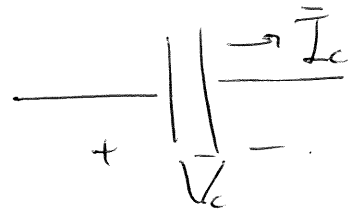


* $\bar{V}_L = j\omega L \bar{I}_L$

C



* $i_C = C \frac{dv_C}{dt}$



* $\bar{I}_C = j\omega C \bar{V}_C$

Note: $\frac{dx(t)}{dt} \rightarrow \bar{X} \times j\omega$

* Impedance: $Z = \frac{\bar{V}}{\bar{I}} \quad [\Omega]$

→ Relationship between the phasor voltage and phasor current.

$R \rightarrow Z_R = R \quad \Omega$

Ohm's Law

$L \rightarrow Z_L = j\omega L \quad \Omega$

for phasor

domain.

$C \rightarrow Z_C = \frac{1}{j\omega C} \quad \Omega$



Note: $\frac{1}{j} = \frac{1}{j} \times \frac{j}{j} = \frac{j}{j^2} = \frac{j}{-1} = -j$

or $\frac{1}{j} = \frac{1}{1\angle 90^\circ} = 1\angle -90^\circ = -j$

$\therefore Z_C = \frac{1}{j\omega C} = -j/\omega C$

* Impedance

$$Z = R + jX \quad (\text{Complex})$$

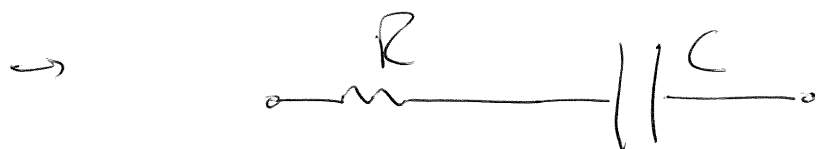
$R \rightarrow$ Resistance (Real part of Z)

$X \rightarrow$ Reactance (Imaginary part of Z)



$$\therefore Z_T = R + j\omega L$$

$$= R + jX_L \quad \text{where } X_L = \omega L$$



$$\therefore Z_T = R + \frac{1}{j\omega C}$$

$$= R + \frac{-j}{\omega C}$$

$$= R + jX_C \quad \text{where } X_C = -\frac{1}{\omega C}$$

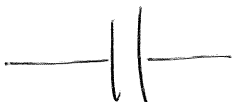
* $f = 0 \text{ Hz}$ and $f \rightarrow \infty \text{ Hz}$

Remember: $\omega = 2\pi f$.

$L //$  $Z = j\omega L$

(DC) If $f = 0 \text{ Hz} \Rightarrow Z = 0 \Omega$ Short circuit

If $f \rightarrow \infty \text{ Hz} \Rightarrow Z \rightarrow \infty \Omega$ Open circuit.

$C //$  $Z = -j/\omega C$

(DC) If $f = 0 \text{ Hz} \Rightarrow Z \rightarrow \infty \Omega$ Open circuit.

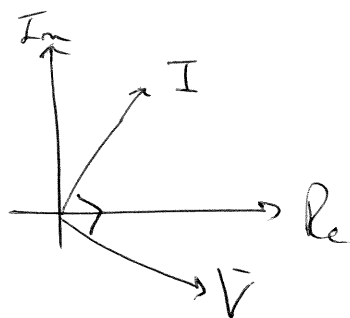
If $f \rightarrow \infty \text{ Hz} \Rightarrow Z \rightarrow 0 \Omega$ Short circuit.

* CIVIL.

→ For a Capacitor, I leads V,
but V leads I for an Inductor.

* C —||— $\bar{I} = j\omega C \bar{V} \quad \left(\frac{\bar{V}}{\bar{I}} = \frac{1}{j\omega C} \right)$

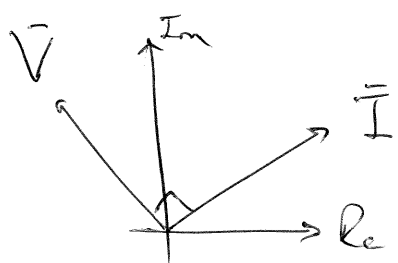
$$\bar{I} = 1 \angle 90^\circ \times \omega C \bar{V}$$



$\bar{I}$ leads $\bar{V}$ by 90°

* L ——— $\bar{V} = j\omega L \bar{I} \quad \left(\frac{\bar{V}}{\bar{I}} = j\omega L \right)$

$$\bar{V} = 1 \angle 90^\circ \omega L \bar{I}$$



$\bar{V}$ leads $\bar{I}$ by 90°

* R ——— $\bar{V} = \bar{I} R$

$\bar{V}$ and $\bar{I}$ are in phase.

* AC Circuit analysis

* KVL: $\sum \bar{V}_{\text{loop}} = 0$

* KCL: $\sum \bar{I}_{\text{out}} = \sum \bar{I}_{\text{in}}$

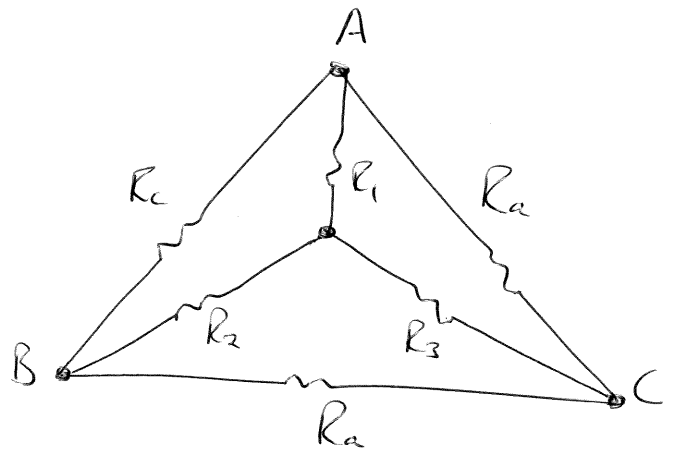
* Series and || impedance. (Z)

* $\bar{V}$ and $\bar{I}$ division.

* Y - Δ

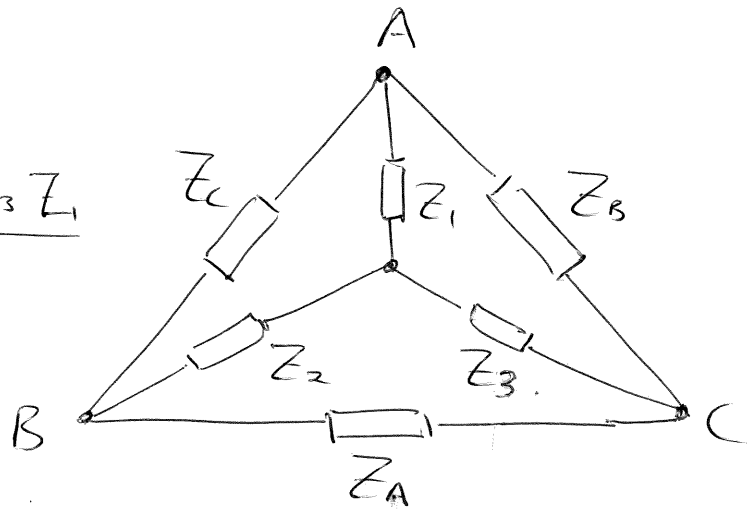
$$\rightarrow R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$



$$\rightarrow Z_A = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_1}$$

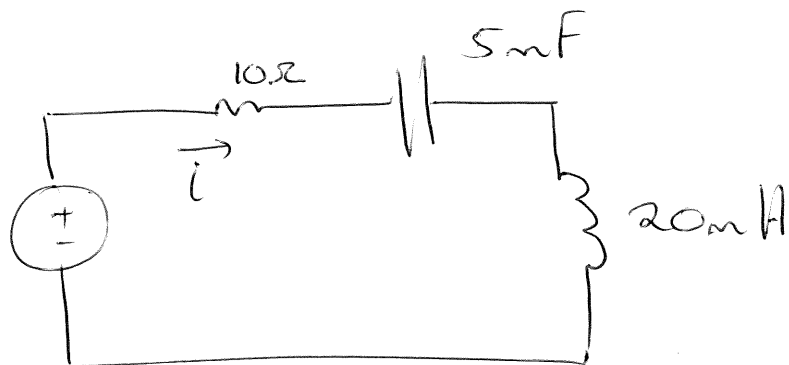
$$Z_1 = \frac{Z_B Z_C}{Z_A + Z_B + Z_C}$$



* Problems.

①

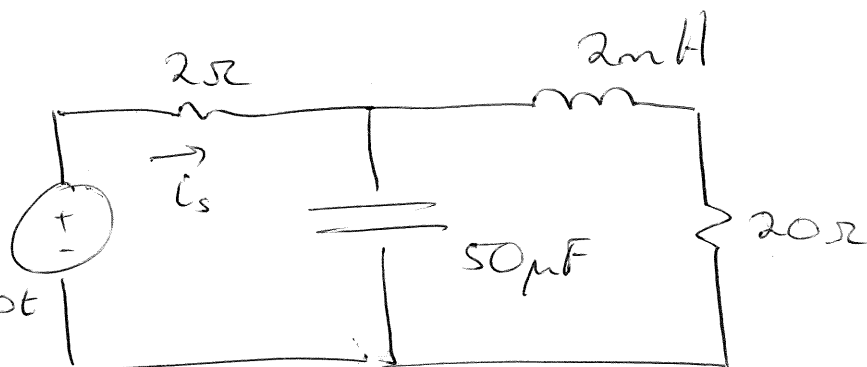
$V_s = 50 \cos 200t$



$i = ?$

②

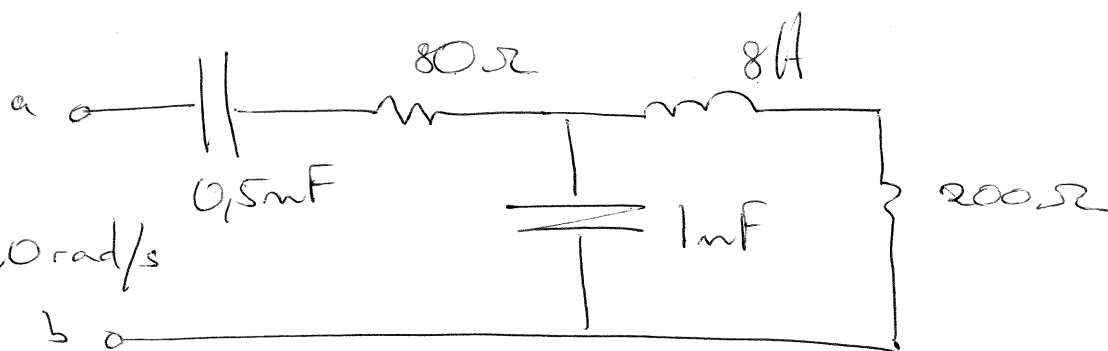
$V_s = 20 \cos 2000t$



$i_s = ?$

③

$\omega = 10 \text{ rad/s}$



$Z_{ab} = ?$